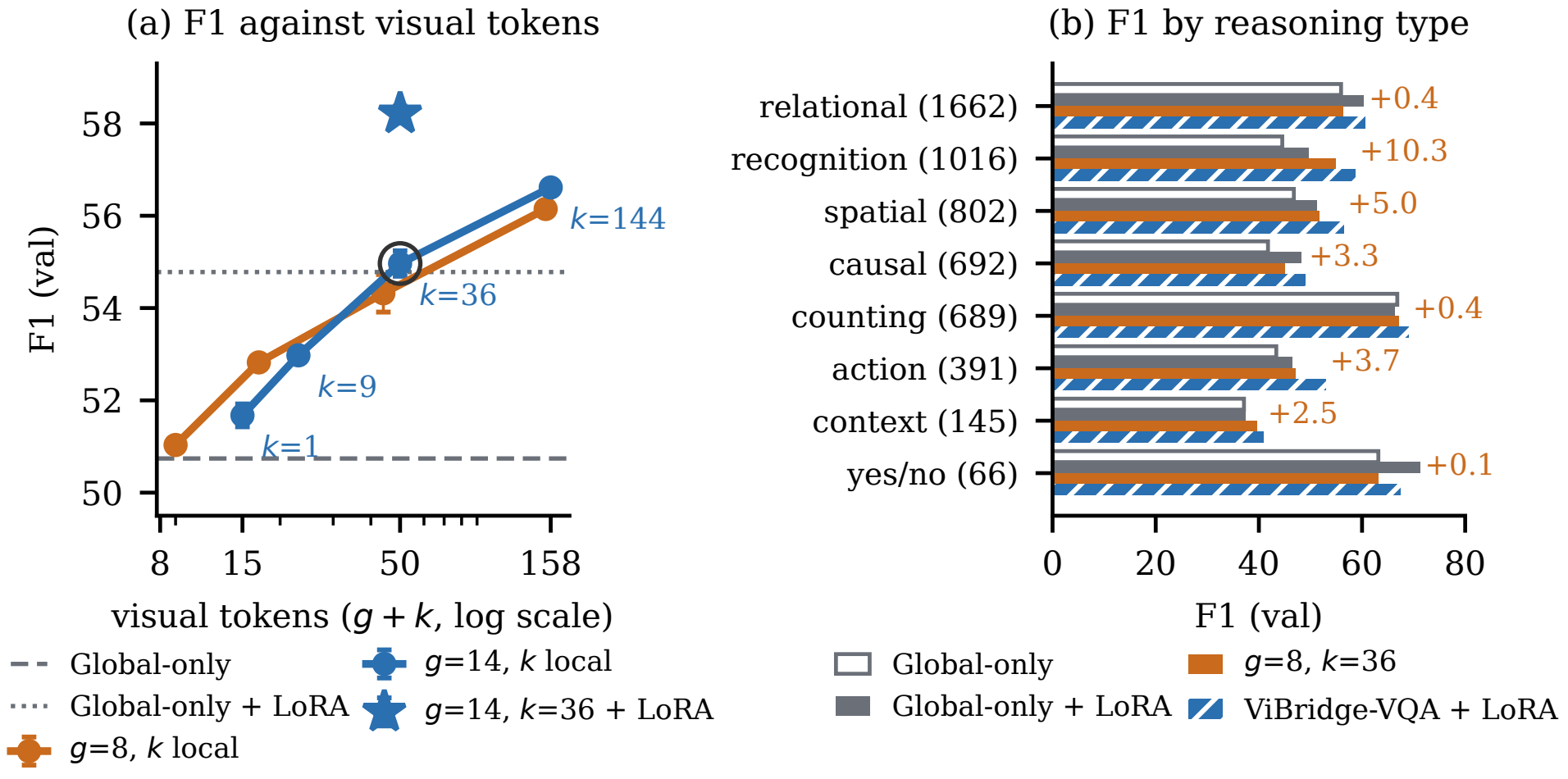
(a) F1 against visual tokens
F1 (val)
58
56
54
52
50
k=144
k=36
k=9
k=1
8
15
50
158
visual tokens (g + k, log scale)
Global-only
Global-only + LoRA
g=8, k local
g=14, k local
g=14, k=36 + LoRA
(b) F1 by reasoning type
relational (1662)
recognition (1016)
spatial (802)
causal (692)
counting (689)
action (391)
context (145)
yes/no (66)
+0.4
+10.3
+5.0
+3.3
+0.4
+3.7
+2.5
+0.1
0
20
40
60
80
F1 (val)
Global-only
Global-only + LoRA
g=8, k=36
ViBridge-VQA + LoRA